FinLens

Multi-Stage Hybrid Search Engine
for Financial Passage Retrieval

57,638	648	FiQA
FINANCIAL PASSAGES	TEST QUERIES	DATASET

ECS736P/U Information Retrieval | Group 2
Artem Zeleniuk · Samuel Edwards · Hardik Mathur · Rayaan Sheikh

PROBLEM

The First-Stage Recall Bottleneck

Cascade retrieval is the standard architecture in modern IR:

Query → **Stage 1: Fast Retrieval** (Top-100 from 57K docs) → **Stage 2: Neural Reranker** (Reorder top-100) → Top-10 Results

Hard ceiling: A relevant document missed at Stage 1 is irrecoverable. No reranker can fix it.

0.559
BM25 RECALL@100 ON FIQA

44%
RELEVANT DOCS LOST

BM25 nDCG@10 across BEIR (Thakur et al., 2021)

DATASET	NDCG@10
HotpotQA (general)	0.633
NQ (general)	0.305
NFCorpus (nutrition)	0.322
FiQA (finance)	**0.236**

Corpora where query and document vocabulary systematically diverge suffer the most. FiQA is the **second-lowest** BM25 baseline across all 18 BEIR datasets: users ask in everyday language, documents use technical financial terminology.

ROOT CAUSE

Vocabulary Mismatch

BM25 treats terms as atomic symbols. Users ask in everyday language; documents use technical financial terminology.

> *"Why would anyone want to pay off their debts in a way other than 'highest interest' first?"*

BM25 Top-5 Results

#	WHAT THE DOCUMENT SAYS	
1	"Pay off the **highest rate debt**"	No
2	"Paying off that **credit card debt**"	No
3	"pay off the **highest interest card**"	No
4	"you should **pay off the highest-interest-rate debt** first"	No
5	"**pay off the debt** with the **highest interest rate** first"	No

All contain the right **words**, but give the **opposite** answer. Only 1 relevant document in the entire BM25 top-10.

APPROACH

Two Competing Pipelines

Pipeline A: Hybrid Retrieval

BM25 | Dense → RRF Fusion → Cross-Encoder

Extends the **retriever**:
adds semantic matching alongside lexical

Pipeline B: Query Expansion

BM25 → Bo1 PRF → BM25 → Cross-Encoder

Extends the **query**:
enriches with related terms, stays lexical

> **CW1 Hypothesis:** *"The encoders will have a contextual blindspot, as they were trained on MS MARCO, not financial text. Query expansion may perform better."*

IMPLEMENTATION

Indexing

BM25: Inverted Index

Maps each term to a posting list of documents

```
"debt"    maps to  [doc_94373, doc_416796, ...]  (3,511 docs)
"snowbal" maps to  [doc_224918, doc_544858, ...]    (67 docs)
"amort"   maps to  [doc_230215, doc_89421, ...]    (146 docs)
```

If a term is not in the query, its posting list is never accessed.

PYTERRIER Build time: under 1 min

Dense: FAISS Vector Index

Each document encoded into a 384-dim vector

```
doc_94373  : [-0.045, 0.022, -0.022, ..., -0.048]
doc_224918 : [-0.011, 0.099, -0.037, ..., -0.016]
...
57,638 vectors x 384 dimensions
```

Exact inner-product search (IndexFlatIP). No approximation loss.

FAISS + ALL-MINILM-L6-V2 Build: approx. 2 min (GPU)

IMPLEMENTATION

Models & Technology Stack

COMPONENT	MODEL / TOOL	WHY CHOSEN
BM25	`PyTerrier` (k_1=1.2, b=0.75)	Standard BEIR baseline, millisecond retrieval
Bi-encoder	`all-MiniLM-L6-v2` (384-dim)	6 layers, <10ms/query, strong OOD transfer
Fusion	Reciprocal Rank Fusion (k=60)	Rank-based merging, no normalisation needed
Cross-encoder	`ms-marco-MiniLM-L-6-v2`	Zero-shot, same MiniLM architecture family
Evaluation	`pytrec_eval`	Recall@100, nDCG@10, MAP, MRR@10

> All neural components used **zero-shot**, without fine-tuning on financial data.

RESULTS

Stage 1: Dense Solves the Bottleneck

CONFIG	RECALL@100	NDCG@10	MAP	MRR@10
BM25	0.559	0.253	0.209	0.321
Dense	**0.706**	**0.369**	**0.310**	**0.454**
Hybrid (RRF)	0.706	0.359	0.303	0.449

+26%

RECALL IMPROVEMENT

+46%

NDCG IMPROVEMENT

> Hybrid (RRF) performs comparably to Dense alone. On this domain, BM25 candidates do not provide additional coverage beyond what Dense already retrieves.

RESULTS

What Dense Found

"Why would anyone want to pay off their debts in a way other than 'highest interest' first?"

3/15
BM25 FOUND

10/15
DENSE FOUND

Documents found ONLY by Dense (sample data from FiQA)

DOCUMENT PASSAGE	WHY BM25 MISSED
"Debt Snowball: pay lowest balance first, purely psychological motivation"	"snowball" absent from query
"Revolving debt vs amortizing loan: flexibility and liquidity"	Financial jargon, no surface overlap
"0% introductory APR: pay off before the higher rate kicks in"	Entirely different framing
"Income fluctuates, sometimes I only have $750"	Discusses practical constraints

PIPELINE B

Query Expansion: Why It Failed

Standard Bo1 (BM25 feedback)

```
Original: debt  highest  interest  pai
          want  wai  first
Added:    card  payment  minimum
          increas  balanc
```

Terms are Porter-stemmed. All added terms are from the same lexical field. Cannot generate "revolv" or "snowbal".

Our hypothesis: Dense feedback

```
Original: debt  highest  interest  pai
          want  wai  first
Added:    (nothing found)
```

Bo1 extracts terms through BM25 inverted index. Bottleneck is in the 2nd pass.

CONFIG	RECALL@100	NDCG@10	CHANGE
BM25	0.559	0.253	
BM25+Bo1	0.563	0.244	-4% (query drift)
DenseBo1	0.560	0.253	no change

Conclusion: In specialised domains, semantic retrieval outperforms lexical expansion.

STAGE 2

Cross-Encoder Reranking

Bi-encoder (Stage 1)

Query and document encoded **independently**.
Compared by cosine similarity.

Fast (ms), but misses fine-grained interactions.

Cross-encoder (Stage 2)

Query and document read **jointly**.
Full cross-attention between all tokens.

Precise, but requires one forward pass per pair.

CONFIG	NDCG@10	MAP	MRR@10	VS BM25
BM25	0.253	0.209	0.321	baseline
Dense	0.369	0.310	0.454	+46%
Dense+CE	**0.380**	**0.302**	**0.450**	**+50%**
BM25+CE	0.347	0.275	0.422	+37%

STAGE 2

Cross-Encoder in Action

"For insurance, why should you refuse $4,000/year for only 10 years and prefer $500/year indefinitely?"

The relevant document discusses breakeven calculation between the two payment plans:

> *"The breakeven amount isn't at 8 years. You calculated how many years of paying $500 it would take to break even with one year of paying $4,000..."*

`Dense: #21` → `CE: #1`

0.000	1.000
NDCG@10 BEFORE CE	NDCG@10 AFTER CE

Cross-encoder understood the mathematical relationship. Bi-encoder, encoding independently, missed it.

APPLICATION

RAG & Demo

Dense+CE Pipeline → Top-5 Documents → GPT-4o-mini → Answer + Citations

Retrieval-Augmented Generation: Every claim in the AI summary is grounded in a retrieved passage with citations.

approx. 1s	approx. 0.5s	approx. 2s
RETRIEVAL	RERANKING	GENERATION

Summary

1. First-stage vocabulary mismatch is the primary bottleneck in domain-specific IR

2. Dense retrieval addresses it through semantic matching (**+26% Recall**)

3. Query expansion cannot escape the lexical paradigm: no recall improvement

4. Best configuration: **Dense + Cross-Encoder** (nDCG@10 = 0.380, **+50%** over BM25)

CW1 hypothesis falsified: Despite MS MARCO training, dense retrieval transfers effectively to the financial domain. Semantic understanding outperforms lexical enrichment.

Thank you